

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

- Fleetguard® Quickdraw Valve - 4 NPTF (1/4-in x 18), Part Number CC2724
- Mini-Gauge Plug - used with Fluid Analysis Kits (1/8-in NPT), Part Number 3318169S and is a Cummins® Filtration part
- Fleetguard® Fluid Vacuum Pump, Part Number CC2802
- Oil Sample Hose, Part Number 5299913
- Probalyzer Cap - used with Fluid Analysis Kits, Part Number 3318168S and is a Cummins® Filtration part

Additional Service Items

- Clean, dry container

General Information

As oil circulates in an engine during normal operation, wear metals and contaminants should accumulate at a steady rate. The amount of contaminants in the oil at the time of sampling generally depends on the length of time since the last oil change. To detect changes in the wear pattern of an engine, the hours the oil has run **must** be recorded when collecting the oil sample.

Information to provide with each sample is listed below.

- Engine model/serial number
- Miles/hours of oil use
- Miles/hours on engine since new or rebuild
- Oil used (brand name, performance category, and viscosity grade)
- Date sample was collected
- Engine application
- Amount of new oil added since previous oil change.

Information to track for assistance in investigating potential issues is listed below.

- Recent engine maintenance
- Analysis of new (unused) oil.

Oil analysis on new (unused) oil **must** be performed to establish a baseline. New (unused) oil analysis samples **must** be taken each time the oil type or oil supplier is changed, or at a minimum, twice a year. Samples **must** be taken from the bulk supply tanks to determine the makeup of the oil and also from the engine to confirm that no contaminants are being introduced by the storage or transfer system.

Make sure the sample is representative of the oil in the engine:

- **Always** collect oil in a clean, dry container.
- Collect a minimum of 118 ml [4 oz] of oil.
- Take the oil sample before adding any new oil to the engine.
- Bring the engine to operating temperature prior to sampling, if possible. This will insure contaminant levels in the oil sample are representative.
- Successive samples **must** be taken in the same manner and from the same location.

Collecting a Sample

Two methods can be used to collect oil samples:

1. Obtain a sample during engine operation by using a pressurized oil sampling port or line (preferred method).
2. Obtain a sample after engine is shut down by using a vacuum to draw oil out of the sump. The sample **must** be obtained as soon as possible after stopping the engine so the oil is still warm and settling has **not** occurred

WARNING

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

WARNING

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

To obtain an oil sample during engine operation, conduct the following.

1. With the engine off, install an oil sampling port or line that allows oil to be collected before the oil is filtered, if possible. If a sampling line is installed, use the shortest line possible.
2. Clean the outside of the valve by wiping with a clean, dry rag.
3. Idle the engine and bring to operating temperature.
4. Purge the sample port or line by allowing a small amount of oil to flow through the valve then discard the purged oil. Purge the entire volume of the sample line and discard or return

the oil to the sump through the oil fill tube.

5. Collect the sample into a clean, dry, bottle from the oil stream being pumped by the idling engine to be sure the sample represents the actual homogeneous oil mixture that is flowing through the engine.

To obtain an oil sample from an engine that is shut down, conduct the following.

1. Use the oil level on the dipstick to determine the length of tube needed to obtain an oil sample.
 - a. Hold a new, clean piece of tubing against the dipstick and mark, on the tube, the location where the dipstick seats.
 - b. Cut the tubing so it reaches 25 to 50 mm (1 to 2 inches) below the oil level in the pan.
2. Insert the sample tube into the dipstick tube so that the mark previously made on the tube is aligned with the top of the opening for the dipstick tube.
3. Use a hand operated vacuum pump (Cummins Filtration™, Part Number CC2802) to collect oil into a clean, dry, bottle.

Note : Do not allow the tube to draw oil from the bottom of the oil pan because excess debris can be collected providing incorrect sampling results.